

Lecture 3: Calculus and Linear Algebra

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Chapter 7: Systems of Linear Equations

Many problems in economics and data science can be modeled as a **system of linear equations**.

In this chapter we recall core facts and **general solution procedures**.

Read Chapter 8 of Werner–Sotskov; Chapter 7 of Simon–Blume.

Exercises 8.5, 8.6, 8.8 (Werner–Sotskov).

Systems of linear equations

Let A be an $m \times n$ matrix, $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ unknown, and $\mathbf{b} \in \mathbb{R}^m$ given. The system

$$A\mathbf{x} = \mathbf{b}$$

has the **vector form**

$$\sum_{j=1}^n x_j \mathbf{a}_j = \mathbf{b},$$

where $\mathbf{a}_1, \dots, \mathbf{a}_n$ are the columns of A .

If $\mathbf{b} = \mathbf{0}$ the system is **homogeneous**; otherwise it is **inhomogeneous**.

An inhomogeneous system can be **inconsistent** (no solution) or **consistent** (at least one solution).

A homogeneous system always has $\mathbf{x} = \mathbf{0}$ as a solution.

Motivating example: ranking the web (PageRank)

Modern search engines rank pages by solving a linear system. Build a row-stochastic transition matrix P where P_{ij} is the probability to move from page i to page j . The long-run visit probabilities $\mathbf{x}$ solve

$$(I - \alpha P^\top) \mathbf{x} = (1 - \alpha) \mathbf{v}, \quad 0 < \alpha < 1,$$

with $\mathbf{v}$ a teleport vector (often uniform).

Mini-web example: pages A, B, C, D with links $A \rightarrow \{B, C\}$, $B \rightarrow C$, $C \rightarrow A$, $D \rightarrow \{A, C\}$. Then

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad \alpha = 0.85, \quad \mathbf{v} = \frac{1}{4}(1, 1, 1, 1)^\top.$$

Solving gives

$$\mathbf{x} \approx (0.380, 0.199, 0.384, 0.038),$$

so C and A rank highest.

Node D is rarely visited because nobody links to it (only teleportation contributes).

Motivating example: deblurring a photo

Phone photos often suffer from motion blur. Blurring is (approximately) linear:

$$b = Ax + \text{noise},$$

where x is the unknown sharp image (vectorized), b is the observed blurred image, and A encodes the blur kernel (convolution).

- **Ideal case:** noise-free, known $A \Rightarrow$ solve the linear system $Ax = b$.
- **Realistic case:** noisy or overdetermined $\Rightarrow$ solve a least-squares problem

$$\min_x \|Ax - b\|^2 + \lambda \|x\|^2,$$

which leads to the normal equations $(A^\top A + \lambda I_n)x = A^\top b$.

Same linear algebra ideas: vector form, consistency, rank, and algorithms to solve large systems.

Rank and existence/uniqueness

Theorem (Column rank = Row rank)

The maximum number of linearly independent columns of A equals the maximum number of linearly independent rows. This common number is the **rank** of A , denoted $\text{rank}(A)$.

Fact: $\text{rank}(A)$ equals the largest order of a **minor** (determinant of a square submatrix) that is nonzero.

Example:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$$

Here $\det(A) = 0$, so $\text{rank}(A) \leq 2$. Since $\det(A_{33}) = \det \begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix} = -2 \neq 0$, we get $\text{rank}(A) = 2$.

Note $2\mathbf{a}_1 - \mathbf{a}_2 = \mathbf{a}_3$, but no single column is a multiple of another. Any two of the three columns form a basis of the column space.

Computing the rank with Gaussian elimination

Elementary row or column operations do not change $\text{rank}(A)$.

$$\begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & -4 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

The **pivot** in each nonzero row is the first nonzero entry from the left.

There are 2 pivots, hence $\text{rank}(A) = 2$.

To find a linear relation among columns, solve the homogeneous system for $(\lambda_1, \lambda_2, \lambda_3)$:

$$\lambda_1 + 2\lambda_2 = 0, \quad -\lambda_2 + \lambda_3 = 0.$$

Thus $(\lambda_1, \lambda_2, \lambda_3) = (-2\lambda_3, \lambda_3, \lambda_3)$ and, e.g., for $\lambda_3 = 1$,

$$-2\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 = \mathbf{0}.$$

Augmented matrix; consistency

The **augmented matrix** of the linear system $A\mathbf{x} = \mathbf{b}$ is $A_{\mathbf{b}} = [A \ \mathbf{b}] \in \mathbb{R}^{m \times (n+1)}$.

Remark: Either $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$ or $\text{rank}(A_{\mathbf{b}}) = \text{rank}(A) + 1$.

Theorem (Rouch–Capelli): $A\mathbf{x} = \mathbf{b}$ is **consistent** $\iff \text{rank}(A_{\mathbf{b}}) = \text{rank}(A)$
(equivalently $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$).

Moreover:

1. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = n$, the solution is **unique**. **Q:** What is it?
2. If $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) < n$, there are **infinitely many** solutions; choose $n - \text{rank}(A)$ variables **free**.

Triangularization (Gaussian elimination)

Theorem

Applying elementary row operations to $A_{\mathbf{b}}$ transforms any system into an equivalent **upper-triangular** one.

This follows essentially from the fact that $A\mathbf{x} = \mathbf{b} \Leftrightarrow A_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0} \Leftrightarrow CA_{\mathbf{b}} \begin{bmatrix} \mathbf{x} \\ -1 \end{bmatrix} = \mathbf{0}$ for any invertible C

$$\begin{array}{rcl} x_1 + x_2 + x_3 = 3 & & x_1 + x_2 + x_3 = 3 \\ x_1 - x_2 + 2x_3 = 2 & \iff & -2x_2 + x_3 = -1 \\ 4x_1 + 6x_2 - x_3 = 9 & & -4x_3 = -4 \end{array}$$

Row ops: $r_2 \leftarrow r_2 - r_1$, $r_3 \leftarrow r_3 - 4r_1$, $r_3 \leftarrow r_3 + r_2$.

We have $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 3 \Rightarrow$ unique solution $\mathbf{x} = (1, 1, 1)$.

A homogeneous example

Consider

$$\begin{aligned}a + b + c + d + e + f &= 0, \\2a + 2b + 2c + 2d - e - f &= 0, \\3a + 3b - c - d - e - f &= 0.\end{aligned}$$

Row-reducing the coefficient matrix gives

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix},$$

so there are 3 pivots, $\text{rank} = 3$, and the solution space has dimension $6 - \text{rank} = 3$:

$$(a, -a, c, -c, e, -e).$$

A basis is $\{(1, -1, 0, 0, 0, 0), (0, 0, 1, -1, 0, 0), (0, 0, 0, 0, 1, -1)\}$.

General solution structure

Theorem: Every solution of $A\mathbf{x} = \mathbf{b}$ can be written as

$$\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p,$$

where $\mathbf{x}_h$ is the **general solution** of $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x}_p$ is any **particular solution** of $A\mathbf{x} = \mathbf{b}$.

Corollary: If $A\mathbf{x} = \mathbf{b}$ is consistent, it has as many solutions as $A\mathbf{x} = \mathbf{0}$.

Example: Three identical equations

$$x_1 + x_2 + x_3 = 1$$

yield $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 1$. Choose two free variables:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \lambda_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}.$$

Two quick examples

1. For $A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 6 & 2 \\ 3 & 2 & 4 \end{pmatrix}$ we found $\text{rank}(A) = 2$. Thus
 $\dim \ker(A) = 3 - 2 = 1$ and

$$\ker(A) = \{\lambda(-2, 1, 1) : \lambda \in \mathbb{R}\}.$$

2. Consider

$$\begin{aligned}x_1 + x_2 + x_3 &= -1, \\x_1 + 2x_2 + 4x_3 &= 2.\end{aligned}$$

Row reduction gives $x_1 + x_2 + x_3 = -1$, $x_2 + 3x_3 = 3$, so
 $\text{rank}(A) = \text{rank}(A_{\mathbf{b}}) = 2$. One free variable:

$$\{\lambda(2, -3, 1) + (-4, 3, 0) : \lambda \in \mathbb{R}\}.$$

Finding a basis for a span

Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix}.$$

Form the 3×3 matrix with these as **columns**:

$$\begin{pmatrix} 2 & -1 & 0 \\ 3 & 0 & -1 \\ 0 & 3 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\text{rank} = 2$ and $3\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$. A convenient basis of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ -1/3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2/3 \end{pmatrix} \right\}$$

(or, equivalently, two of the original vectors that are independent).

Standing on the shoulders of giants



“If I have seen further, it is by standing on the shoulders of giants.” **Isaac Newton (1675)**.

Some of the giants of linear/matrix algebra:

- Gauss (Gaussian elimination, least squares, determinants)
- Cauchy (CauchySchwarz, eigenvalues, determinants)
- Laplace (Laplace expansion of determinants)
- Cayley (CayleyHamilton theorem, matrix theory)
- Sylvester (matrix rank, invariant theory)
- Jordan (Jordan canonical form)
- Frobenius (matrix factorizations, linear algebra foundations)

(pictured here as the base of a *castell*)

Chapter 8: Eigenvalues and Quadratic Forms

Eigenvalues/eigenvectors appear in growth models, Markov chains, PCA, and stability of dynamical systems.

Quadratic forms drive optimization via second-order conditions.

Read Chapter 10 of Werner–Sotskov (optional: Simon–Blume, Ch. 7).

Exercises 10.2; 10.5 A,B,C; 10.6(a,b).

Eigenvalues and eigenvectors: migration example

Consider a population split between two regions: city (x_t^C) and countryside (x_t^R) at time t . Suppose each year a fraction of people move between the two regions:

$$\begin{pmatrix} x_{t+1}^C \\ x_{t+1}^R \end{pmatrix} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix} \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix} = A \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix}.$$

Here, 0.8 means that 80% of city dwellers stay in the city, while 0.4 means that 40% of rural dwellers move to the city in the next period.

If the **urban/rural ratio stabilizes**, the next state must be a constant multiple of the current state:

$$\begin{pmatrix} x_{t+1}^C \\ x_{t+1}^R \end{pmatrix} = \lambda \begin{pmatrix} x_t^C \\ x_t^R \end{pmatrix}.$$

The long-run stable distribution of the population is given by an **eigenvector** of A , and λ is the associated **eigenvalue** describing the growth rate.

Eigenvalues and eigenvectors: definitions

For $A \in \mathbb{R}^{n \times n}$, $\lambda \in \mathbb{R}$ is an **eigenvalue** if there exists $\mathbf{x} \neq \mathbf{0}$ with

$$A\mathbf{x} = \lambda\mathbf{x} \iff (A - \lambda I_n)\mathbf{x} = \mathbf{0}.$$

Then $\mathbf{x}$ is an **eigenvector** for λ .

Characterization:

$$\lambda \text{ eigenvalue} \iff \text{rank}(A - \lambda I_n) < n \iff \det(A - \lambda I_n) = 0.$$

The **characteristic polynomial** $P(\lambda) = \det(A - \lambda I_n)$ has degree n ; the real roots (at most n) are the eigenvalues.

Procedure:

1. Solve $\det(A - \lambda I_n) = 0$ to get eigenvalues.
2. For each λ , solve $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ for eigenvectors.

Cohort-mix example (continued)

With $A = \begin{pmatrix} 0.8 & 0.4 \\ 0.3 & 0.9 \end{pmatrix}$,

$$(0.8 - \lambda)(0.9 - \lambda) - 0.12 = 0 \iff \lambda^2 - 1.7\lambda + 0.6 = 0.$$

The solutions are $\lambda_1 = 1.2$ and $\lambda_2 = 0.5$.

For $\lambda_1 = 1.2$, eigenvectors are multiples of $(1, 1)$. For $\lambda_2 = 0.5$, eigenvectors are multiples of $(1, -\frac{3}{4})$.

For a population-growth interpretation we keep the dominant eigenpair: equal cohort proportions grow by 20% per step.

Basic spectral facts

If $A \in \mathbb{R}^{n \times n}$,

1. Eigenvectors associated with distinct eigenvalues are **linearly independent**.
2. If A is **symmetric**, then all eigenvalues are real and eigenvectors for distinct eigenvalues are **orthogonal** (Spectral Theorem).

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then $P(\lambda) = \lambda^2 - 2\lambda - 3$, so $\lambda_1 = 3$, $\lambda_2 = -1$ with eigenvectors along $(1, 1)$ and $(-1, 1)$, respectively.

Take the spectral decomposition $A = U\Lambda U^T$. Show columns of U are the eigenvectors.

Repeated eigenvalues

Consider

$$A = \begin{pmatrix} 0 & -1 & 1 \\ -7 & 0 & 5 \\ -5 & -2 & 5 \end{pmatrix}.$$

The characteristic polynomial is

$$p_A(\lambda) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 2)^2(\lambda - 1).$$

Hence the eigenvalues are $\lambda = 2$ (algebraic multiplicity 2) and $\lambda = 1$ (algebraic multiplicity 1).

Eigenspaces:

$$E_2 = \ker(A - 2I) = \text{span}\{(1, -1, 1)\}, \quad E_1 = \ker(A - I) = \text{span}\{(2, 1, 3)\}.$$

Since $\dim E_2 = 1 < 2$, A is **defective** (not diagonalizable).

Application: Principal Component Analysis (PCA)

We observe n data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$.

Step 1 (Centering): Arrange the data in a matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows $\mathbf{x}_1, \dots, \mathbf{x}_n$. Define the centering matrix $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ and then the **centered data** is $H\mathbf{X}$.

Step 2 (Sample covariance):

$$S = \frac{1}{n}(H\mathbf{X})^\top(H\mathbf{X}).$$

Step 3 (Spectral decomposition):

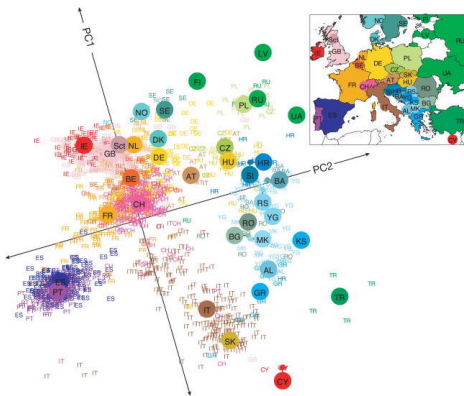
$$S = U\Lambda U^\top \quad \text{and define } \mathbf{Y} = H\mathbf{X}U \quad (\text{rotate the centered data}).$$

The sample covariance of $\mathbf{Y}$ is $\frac{1}{n}\mathbf{Y}^\top\mathbf{Y} = U^\top S U = \Lambda$ (diagonal!).

The columns of U are **principal directions** and diagonal entries of Λ are the **explained variances**.

Sorting eigenvalues in decreasing order ranks components by variance captured.

PCA: why it matters



Dimensionality reduction: compress high-dimensional data into a few directions that preserve most variability.

Noise filtering: small eigenvalues $\rightarrow$ directions with little signal.

Visualization: projecting onto first 2 PCs often reveals hidden structure.

Nice example: genetic variation in Europeans plotting individuals on the first two PCs of SNP data *recovers the geographic map of Europe!*

Numerical toy example

Covariance matrix:

$$S = \begin{pmatrix} 72.9 & 87.5 & 6.875 \\ 87.5 & 108.3 & 8.75 \\ 6.875 & 8.75 & 0.73 \end{pmatrix}.$$

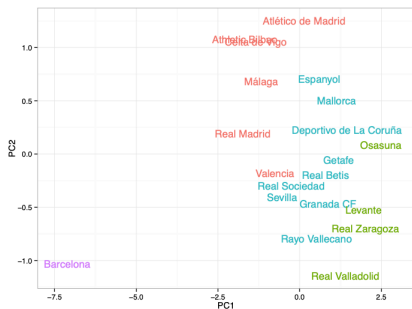
Spectral decomposition:

$$S = U\Lambda U^{\top}, \quad \Lambda = \text{diag}(180, 1.38, 0.005).$$

⇒ The first component explains > 99% of the variance; the last can be safely ignored.

PCA & Unique football styles

“Mille viae ducunt Barcinonem” (A thousand roads lead to Barcelona)



“Searching for a Unique Style in Soccer” (Gyarmati, Kwak & Rodriguez, 2014), studies sequential pass patterns in a team’s passing.

They defined “flow motifs” as statically significant pass subsequences (e.g., $A \rightarrow B \rightarrow C \rightarrow A$).

Applied motif analysis to compare styles across teams and leagues. The data are projected to the first two principal components.

Quadratic forms

A **quadratic form** is $Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = \sum_{i,j=1}^n a_{ij} x_i x_j$ with $A \in \mathbb{R}^{n \times n}$ symmetric.

Remark: Since the coefficient of $x_i x_j$ is $a_{ij} + a_{ji}$, every quadratic form has a unique symmetric representative (replace A by $\frac{1}{2}(A + A^\top)$).

Example: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 5/2 \\ 5/2 & 4 \end{pmatrix}$ define the same

$$Q(x, y) = x^2 + 5xy + 4y^2.$$

Signs of quadratic forms

For symmetric A :

1. Positive definite: $\mathbf{x}^\top A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
2. Negative definite: $\mathbf{x}^\top A \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$.
3. Positive semidefinite: $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$.
4. Negative semidefinite: $\mathbf{x}^\top A \mathbf{x} \leq 0$ for all $\mathbf{x}$.
5. Indefinite: takes both positive and negative values.

Example:

$$\mathbf{x}^\top \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{x} = (x - y)^2 \geq 0$$

so the matrix is positive semidefinite.

Eigenvalue and minor tests

Spectral test (symmetric A):

- PD $\iff$ all eigenvalues > 0 ; ND $\iff$ all < 0 ;
- PSD $\iff$ all ≥ 0 ; NSD $\iff$ all ≤ 0 ;
- Indefinite $\iff$ eigenvalues of different signs.

Sylvester's criterion (leading principal minors D_k):

- PD $\iff D_k > 0$ for $k = 1, \dots, n$;
- ND $\iff (-1)^k D_k > 0$ for $k = 1, \dots, n$.

Example: $A = \begin{pmatrix} -3 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$ has $D_1 = -3$, $D_2 = 5$, $D_3 = -25 \Rightarrow$
negative definite.

Examples

1. $A = \begin{pmatrix} 1 & 4 & 6 \\ 4 & 2 & 1 \\ 6 & 1 & 6 \end{pmatrix}$: $D_1 = 1$, $D_2 = -14$, $D_3 = -109 \Rightarrow$ indefinite.

2. $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}$: $D_1 = 3$, $D_2 = 5$, $D_3 = 12 \Rightarrow$ positive definite.

3. The symmetric matrix of $Q(x, y, z) = xy + yz$ is indefinite since $D_1 = D_3 = 0$ and $D_2 = -\frac{1}{4}$.